Solution Requirements

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

1: Functional Requirements (FR) :

Functional Requirements define **what the system should do**. They describe the core functionalities and services that the application must provide to achieve its intended purpose. In this project, functional requirements include collecting applicant details, validating user inputs, processing the information through the trained Random Forest machine learning model, generating accurate approval predictions, and displaying the prediction results through a user-friendly web interface. These requirements ensure that every component of the system works together to deliver reliable credit card approval predictions.

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the application through the web interface. (Optional login can be added in future versions.)	High
2	Authorization Levels	The system provides appropriate access to users. Customers can submit application details, while administrators can manage the machine learning model and application settings.	Medium
3	External Interfaces	The system interacts with the trained Random Forest model (`credit_card_approval_model.pkl`) and accepts user inputs through the Flask web application.	High
4	Transaction Processing	The system collects applicant information, validates the data, processes it using the trained model, and instantly returns the prediction result.	High
5	Reporting	The application displays the prediction result (Approved or Rejected). Future versions can generate reports and prediction history.	Medium

6	Business Rules	The prediction must be based on applicant information such as age, income, education, occupation, employment, family status, and other financial attributes. Input data must follow the same preprocessing and encoding used during model training.	High
7	Compliance to Laws or Regulations	Applicant information should be handled securely and used only for prediction purposes while maintaining data privacy and confidentiality.	High
8	Other	The application should validate user inputs before prediction and display meaningful error messages if invalid or incomplete information is entered.	Medium

2: Non-Functional Requirements (NFR) :

Non-Functional Requirements define **how the system should perform**. They focus on the quality attributes of the application rather than its specific functionalities. For the Credit Card Approval Prediction system, non-functional requirements include fast prediction response time, high prediction accuracy, data security, system reliability, ease of use, scalability, and maintainability. These requirements ensure that the application is efficient, secure, user-friendly, and capable of handling future enhancements and increased user demand.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate prediction results quickly after the user submits the form.	Prediction generated within **2–3 seconds**.
2	Scalability	The system should support multiple users and allow future expansion with larger datasets or additional prediction models.	Support **100+ simultaneous requests** without significant performance degradation.
3	Security & Data Privacy	Applicant information should be processed securely without exposing sensitive personal data.	Secure input validation and safe handling of user information.
4	Reliability & Availability	The system should provide consistent and accurate predictions with minimal downtime.	**99% availability** during normal operation.
5	Usability & Accessibility	The web application should provide a simple, intuitive, and responsive interface that is easy to use.	Easy navigation with clear labels and responsive design.
6	Maintainability	The application should be	Modular code structure with

		modular so that the machine learning model, frontend, or backend can be updated independently.	proper documentation and comments.